

Separable Domains Give All-Codomain Projections in $\text{Lip}_0(X, Y)$

Automatic math-discovery packet

June 29, 2026

Source question

The source paper is Anil Kumar Karn and Arindam Mandal, *Position of $L(X, Y)$ in $\text{Lip}_0(X, Y)$* , arXiv:2506.09324. Question 1.1 asks whether $L(X, Y)$ is complemented in $\text{Lip}_0(X, Y)$.

POSITION OF $L(X, Y)$ IN $Lip_0(X, Y)$

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ABSTRACT. We prove that $L(X, Y)$ is complemented in $Lip_0(X, Y)$ (via a norm-one projection) provided that Y is a dual space. Next, we introduce a vector-valued Lipschitz-free space $F_Y(X)$, a linear contraction $\beta_X^Y : F_Y(X) \rightarrow Y$ and prove that the quotient space $Lip_0(X, Y)/L(X, Y)$ is isometrically isomorphic to $L(\ker(\beta_X^Y), Y)$ whenever Y is injective. We also consider a Y -valued duality pairing between $Lip_0(X, Y)$ and $F_Y(X)$ and obtain a necessary and sufficient condition for a Lipschitz map to be linear. As an application, we describe $\ker(\beta_X^{\mathbb{R}})$ as the pre-dual of the quotient space $L^\infty(\mathbb{R})/\bar{\mathbb{R}}$ where $\bar{\mathbb{R}}$ is the set of all constant maps on $\mathbb{R}$.

1. INTRODUCTION AND PRELIMINARIES

A mapping $f : X \rightarrow Y$ between two real Banach spaces X and Y is called Lipschitz if there exists $k > 0$ in $\mathbb{R}$ such that $\|f(x_1) - f(x_2)\| \leq k\|x_1 - x_2\|$ for all $x_1, x_2 \in X$. The set of all such mappings is denoted by $Lip(X, Y)$, which forms a vector space with respect to pointwise addition and scalar multiplication. In the study of Lipschitz mappings, special attention is given to those mappings that send the 0 to 0. The set of such mappings is denoted by $Lip_0(X, Y)$. That is, $Lip_0(X, Y) = \{f \in Lip(X, Y) : f(0) = 0\}$. Further $Lip_0(X, Y)$ is a Banach space with the norm

$$Lip(f) = \sup \left\{ \frac{\|f(x) - f(y)\|}{\|x - y\|} : x, y \in X, x \neq y \right\} \text{ for any } f \in Lip_0(X, Y).$$

The space $Lip_0(X, Y)$ is often regarded as a natural non-linear generalization of $L(X, Y)$, the space of bounded linear operators between X and Y . It is known that $L(X, Y)$ is a subspace of $Lip_0(X, Y)$. The space $L(X, Y)$ carries the usual operator norm,

$$\|T\| = \sup\{\|Tx\| : x \in X \text{ and } \|x\| \leq 1\}$$

for any $T \in L(X, Y)$. We have $\|T\| = Lip(T)$ for all $T \in L(X, Y)$ so that $L(X, Y)$ is a closed subspace of $Lip_0(X, Y)$. This leads to the following question:

Question 1.1. *Is $L(X, Y)$ complemented in $Lip_0(X, Y)$?*

The answer to this question is known to be affirmative in the case $Y = \mathbb{R}$. In fact, by a result due to Lindenstrauss ([5, Theorem 2]) we know that there exists a (norm one)

2020 *Mathematics Subject Classification.* Primary 46B20; Secondary 46B03.

Key words and phrases. Invariant mean, Vector-valued duality, Generalized Lipschitz-free space.

Second author is supported by a research fellowship from the Department of Atomic Energy (DAE), Government of India.

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Figure 1: Question 1.1 asks whether $L(X, Y)$ is complemented in $Lip_0(X, Y)$.

Source inputs

Karn–Mandal define $F_Y(X)$ as the closed linear span of the evaluation maps $\delta_x^Y : \text{Lip}_0(X, Y) \rightarrow Y$, and define a contractive linear barycenter map

$$\beta_X^Y : F_Y(X) \longrightarrow X, \quad \beta_X^Y(\delta_x^Y) = x.$$

They prove that

$$\Delta_X^Y : L(F_Y(X), Y) \longrightarrow \text{Lip}_0(X, Y), \quad \Delta_X^Y(A) = A \circ \delta_X^Y,$$

is a surjective linear isometry, and that

$$L(X, Y) = \diamond \ker(\beta_X^Y).$$

Equivalently, under Δ_X^Y , the subspace $L(X, Y)$ corresponds to the operators $A : F_Y(X) \rightarrow Y$ that vanish on $\ker(\beta_X^Y)$, i.e. to the operators factoring through β_X^Y .

We also use the Godefroy–Kalton lifting theorem: if X is separable, then the usual scalar Lipschitz-free barycenter map

$$\beta_X : \mathcal{F}(X) \longrightarrow X$$

has a norm-one bounded linear right inverse. We cite this in the form recorded in Ferenczi–Kaufmann–Pernecka, *Equivariant liftings in Lipschitz-free spaces*, arXiv:2501.06984, Theorem 2.

Proof intuition

The source paper’s generalized space $F_Y(X)$ looks Y -valued, but its underlying molecule norm is actually the scalar Lipschitz-free norm. Testing a Y -valued Lipschitz function against Y^* gives the upper bound by the scalar free norm, while scalar Lipschitz functions multiplied by a fixed unit vector of Y give the reverse inequality.

Thus any linear lifting of the scalar barycenter map $\beta_X : \mathcal{F}(X) \rightarrow X$ becomes a linear lifting of $\beta_X^Y : F_Y(X) \rightarrow X$. A linear lifting splits

$$0 \longrightarrow \ker(\beta_X^Y) \longrightarrow F_Y(X) \xrightarrow{\beta_X^Y} X \longrightarrow 0,$$

and this split gives a projection on operator spaces. Transferring it through Δ_X^Y gives a projection from $\text{Lip}_0(X, Y)$ onto $L(X, Y)$.

The scalar free space is already present

Lemma 1. *Let $Y \neq \{0\}$. The linear map initially defined on molecules by*

$$U \left(\sum_{i=1}^n a_i \delta_{x_i} \right) = \sum_{i=1}^n a_i \delta_{x_i}^Y$$

extends to a surjective linear isometry

$$U : \mathcal{F}(X) \longrightarrow F_Y(X).$$

Moreover $\beta_X^Y U = \beta_X$.

Proof. For a finite molecule $m = \sum_i a_i \delta_{x_i}$, its $F_Y(X)$ -norm is

$$\|Um\|_{F_Y(X)} = \sup_{\text{Lip}(f) \leq 1} \left\| \sum_i a_i f(x_i) \right\|,$$

where f ranges over $\text{Lip}_0(X, Y)$. For every $y^* \in B_{Y^*}$, $y^* \circ f \in B_{\text{Lip}_0(X)}$. Hence

$$\left\| \sum_i a_i f(x_i) \right\| = \sup_{y^* \in B_{Y^*}} \left| \sum_i a_i y^*(f(x_i)) \right| \leq \|m\|_{\mathcal{F}(X)}.$$

Thus $\|Um\|_{F_Y(X)} \leq \|m\|_{\mathcal{F}(X)}$.

Conversely, choose $y_0 \in Y$ with $\|y_0\| = 1$. For every $g \in B_{\text{Lip}_0(X)}$, the map $f_g(x) = g(x)y_0$ belongs to $B_{\text{Lip}_0(X, Y)}$, and

$$\left\| \sum_i a_i f_g(x_i) \right\| = \left| \sum_i a_i g(x_i) \right|.$$

Taking the supremum over g gives $\|Um\|_{F_Y(X)} \geq \|m\|_{\mathcal{F}(X)}$. Therefore U is an isometry on the molecule span, and it extends to an isometry $\mathcal{F}(X) \rightarrow F_Y(X)$. Its range is dense by definition of $F_Y(X)$, and closed because it is isometric, so it is onto. Finally, $\beta_X^Y U(\delta_x) = \beta_X^Y(\delta_x^Y) = x = \beta_X(\delta_x)$, and the identity extends by continuity. $\square$

Lifting-property projection

Theorem 1. *Suppose the scalar barycenter map $\beta_X : \mathcal{F}(X) \rightarrow X$ has a bounded linear right inverse $S : X \rightarrow \mathcal{F}(X)$, with $\beta_X S = I_X$. Then for every Banach space Y , the subspace $L(X, Y)$ is complemented in $\text{Lip}_0(X, Y)$. More precisely, if $Y \neq \{0\}$, there is a projection of norm at most $\|S\|$; if $Y = \{0\}$, the assertion is trivial.*

Proof. Assume $Y \neq \{0\}$. Let $U : \mathcal{F}(X) \rightarrow F_Y(X)$ be the isometry of the lemma and put $S_Y = US : X \rightarrow F_Y(X)$. Then

$$\beta_X^Y S_Y = \beta_X^Y U S = \beta_X S = I_X.$$

Define

$$Q : L(F_Y(X), Y) \longrightarrow L(F_Y(X), Y), \quad Q(A) = A S_Y \beta_X^Y.$$

Since $\|\beta_X^Y\| \leq 1$, we have $\|Q\| \leq \|S_Y\| \leq \|S\|$. Also $Q^2 = Q$, because $\beta_X^Y S_Y = I_X$.

The range of Q consists of operators that factor through β_X^Y , hence vanish on $\ker(\beta_X^Y)$. Conversely, if $A \in L(F_Y(X), Y)$ vanishes on $\ker(\beta_X^Y)$, then A factors through β_X^Y , say $A = B \beta_X^Y$ for some $B \in L(X, Y)$, and

$$Q(A) = B \beta_X^Y S_Y \beta_X^Y = B \beta_X^Y = A.$$

Thus Q is a projection onto the operators corresponding, under the Karn–Mandal isometry Δ_X^Y , to $L(X, Y)$. Therefore

$$\Pi = \Delta_X^Y Q (\Delta_X^Y)^{-1}$$

is a bounded projection from $\text{Lip}_0(X, Y)$ onto $L(X, Y)$, with $\|\Pi\| \leq \|S\|$. $\square$

Corollary 1. *If X is separable, then for every Banach space Y , $L(X, Y)$ is one-complemented in $\text{Lip}_0(X, Y)$.*

Proof. By the Godefroy–Kalton lifting theorem, separable X has a norm-one linear right inverse $S : X \rightarrow \mathcal{F}(X)$ for β_X . Apply the theorem. $\square$

Limitations and novelty check

This is not a full answer to Question 1.1 for arbitrary X and Y . The proof uses the scalar lifting property of X , and Ferenczi–Kaufmann–Pernecka recall that some Banach spaces fail this lifting property, citing Godefroy–Kalton’s examples. Thus this route cannot simply be pushed to all domains.

The result is nevertheless independent of the previous codomain-duality route: it assumes nothing about Y . Cheap run-index searches found only the previous arXiv:2506.09324 bidual-complemented-codomain packet and an adjacent two-Lipschitz tensor-splitting packet, but no exact packet for the one-variable $\text{Lip}_0(X, Y)$ separable-domain/all-codomain theorem.

Human-review recommendation

Review as a likely valid partial result. The key points to check are: first, the norm equality between $F_Y(X)$ and the scalar $\mathcal{F}(X)$; second, that the projection $A \mapsto AS_Y\beta_X^Y$ has range exactly the operators vanishing on $\ker(\beta_X^Y)$, which the source paper identifies with $L(X, Y)$ after applying Δ_X^Y .

References

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